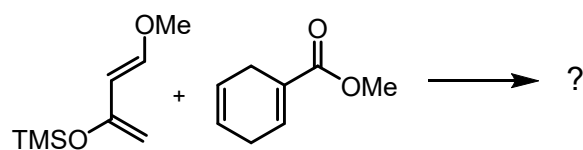
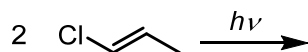


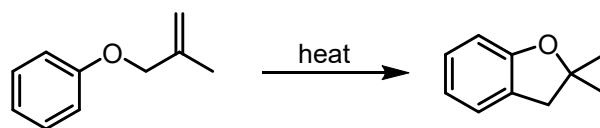
1. Predict the product of this Diels-Alder reaction and explain the selectivity.



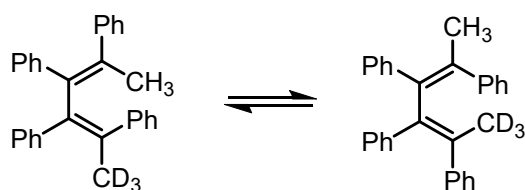
2. Give the possible products with relative stereochemistry of the following reaction.



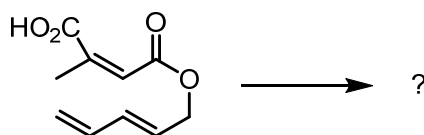
3. Suggest a mechanism for this unexpected reaction.



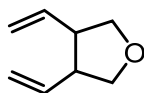
4. The following thermal isomerization occurs under relatively mild conditions. Identify the pericyclic reactions involved, and show how the rearrangement occurred.



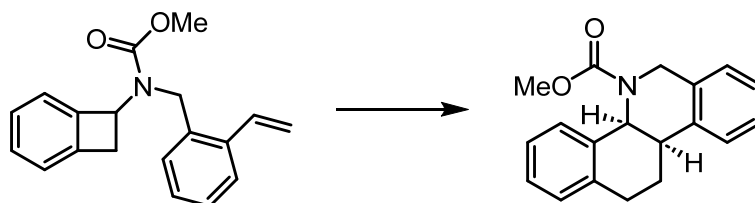
5. Predict the product of this Diels-Alder reaction and explain the selectivity.



6. Using pericyclic reaction as the key reaction to synthesize this compound from the starting materials containing 3 carbon atoms or less.



7. Suggest a mechanism for this reaction.



8. Propose a mechanism to produce Honokiol and Magnolol from compound **A**.

